

Summer Math Adventure

W2
D1

Week 2 · Day 1 | Fractions: Meaning & Equivalent Fractions

Name: _____

Date: _____

FRACTIONS

Fraction = part of a whole. $\frac{\text{numerator}}{\text{denominator}}$ → numerator = how many parts you have, denominator = total equal parts.

Equivalent fractions: multiply OR divide numerator AND denominator by the same number.

$$\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$$

Part A – Shade & Label

Shade the correct fraction of each bar. Then write an equivalent fraction by doubling.

1. Shade $\frac{3}{4}$ — then find an equivalent fraction with denominator 8.

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Equivalent: $\frac{3}{4} = \frac{\boxed{}}{8}$

2. Shade $\frac{2}{5}$ — then find an equivalent fraction with denominator 15.

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Equivalent: $\frac{2}{5} = \frac{\boxed{}}{15}$

Part B – Find the Missing Number

3. $\frac{3}{5} = \frac{\boxed{}}{20}$

4. $\frac{4}{6} = \frac{2}{\boxed{}}$

5. $\frac{5}{8} = \frac{\boxed{}}{40}$

6. $\frac{18}{24} = \frac{3}{\boxed{}}$

Part C – Simplest Form

Divide numerator and denominator by their GCF (Greatest Common Factor).

7. $\frac{8}{12} = \frac{\boxed{}}{\boxed{}}$

8. $\frac{15}{20} = \frac{\boxed{}}{\boxed{}}$

9. $\frac{24}{36} = \frac{\boxed{}}{\boxed{}}$

10. $\frac{30}{45} = \frac{\boxed{}}{\boxed{}}$

Part D – Compare Fractions

Write $<$, $>$, or $=$. First convert to a common denominator.

11. $\frac{3}{4} \boxed{} \frac{5}{8}$

12. $\frac{2}{3} \boxed{} \frac{3}{5}$

Common denom: _____

Common denom: _____

Summer Math Adventure

W2
D2

Week 2 · Day 2 | Adding & Subtracting Fractions with Unlike Denominators

Name: _____

Date: _____

FRACTIONS: ADD & SUBTRACT

To add or subtract fractions with different denominators:

Step 1: Find the **LCD** (Least Common Denominator)

Step 2: Convert each fraction to an equivalent fraction using the LCD

Step 3: Add or subtract the numerators. Keep the denominator.

Step 4: Simplify if needed.

Example: $\frac{1}{3} + \frac{1}{4} \rightarrow \text{LCD} = 12 \rightarrow \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

Part A – Find the LCD First

1. Denominators: 4 and 6 LCD =

2. Denominators: 3 and 8 LCD =

3. Denominators: 5 and 6 LCD =

4. Denominators: 4 and 9 LCD =

Part B – Add the Fractions

Show the LCD step. Simplify your answer.

5. $\frac{1}{2} + \frac{1}{5} =$

LCD:

$$\frac{\boxed{}}{\boxed{}} + \frac{\boxed{}}{\boxed{}} =$$

6. $\frac{2}{3} + \frac{1}{4} =$

LCD:

$$\frac{\boxed{}}{\boxed{}} + \frac{\boxed{}}{\boxed{}} =$$

7. $\frac{3}{4} + \frac{2}{5} =$

LCD:

=

8. $\frac{5}{6} + \frac{3}{8} =$

LCD:

=

Part C – Subtract the Fractions

9. $\frac{3}{4} - \frac{1}{3} =$

10. $\frac{5}{6} - \frac{1}{4} =$

Summer Math Adventure

W2
D3

Week 2 · Day 3 | Mixed Numbers: Converting, Adding & Subtracting

Name: _____

Date: _____

MIXED NUMBERS

Mixed Number: whole number + fraction (e.g., $3\frac{1}{2}$)

Convert to improper fraction: multiply whole $\times$ denominator, add numerator, keep denominator.

$$3\frac{1}{2} \rightarrow (3 \times 2) + 1 = 7 \rightarrow \frac{7}{2}$$

Convert to mixed number: divide. Quotient = whole, remainder = new numerator.

$$\frac{11}{4} \rightarrow 11 \div 4 = 2 \text{ R } 3 \rightarrow 2\frac{3}{4}$$

Part A – Convert Between Forms

Mixed $\rightarrow$ Improper:

1. $2\frac{3}{4} = \frac{\boxed{}}{\boxed{}}$

2. $5\frac{2}{3} = \frac{\boxed{}}{\boxed{}}$

3. $4\frac{5}{6} = \frac{\boxed{}}{\boxed{}}$

Improper $\rightarrow$ Mixed:

4. $\frac{17}{5} =$

5. $\frac{29}{4} =$

6. $\frac{41}{6} =$

Part B – Add Mixed Numbers

Add whole numbers + add fractions. Convert to LCD first. Regroup if fraction ≥ 1 .

7. $3\frac{1}{2} + 2\frac{1}{4} =$

8. $4\frac{2}{3} + 1\frac{3}{4} =$

9. $2\frac{5}{6} + 3\frac{3}{4} =$

10. $6\frac{7}{8} + 2\frac{3}{4} =$

Part C – Subtract Mixed Numbers

Summer Math Adventure

W2
D4

Week 2 · Day 4 | Multiplying Fractions & Fractions of Whole Numbers

Name: _____

Date: _____

MULTIPLYING FRACTIONS

Multiply fractions: top $\times$ top, bottom $\times$ bottom.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Fraction of a whole number: multiply the whole number by the fraction.

$$\frac{3}{4} \text{ of } 24 = \frac{3}{4} \times 24 = \frac{3 \times 24}{4} = \frac{72}{4} = 18$$

Tip: Cross-cancel before multiplying to keep numbers small!

Part A – Multiply Fractions

1. $\frac{1}{2} \times \frac{3}{5} =$

2. $\frac{3}{4} \times \frac{2}{9} =$

3. $\frac{5}{6} \times \frac{3}{10} =$

4. $\frac{4}{7} \times \frac{7}{8} =$

5. $\frac{2}{3} \times \frac{6}{7} =$

6. $\frac{5}{8} \times \frac{4}{5} =$

Part B – Fraction of a Whole Number

7. $\frac{3}{4}$ of 32 =

8. $\frac{2}{5}$ of 45 =

9. $\frac{5}{6}$ of 54 =

10. $\frac{7}{8}$ of 64 =

Part C – Word Problems

11. A hiking trail is $4\frac{1}{2}$ miles. Sofia walked $\frac{2}{3}$ of the trail. How far did she walk?

Answer:

12. A class of 30 students: $\frac{3}{5}$ chose soccer, $\frac{1}{6}$ chose swimming, and the rest chose basketball. How many chose each sport?

Summer Math Adventure

W2
D5

Week 2 · Day 5 | Fractions on Number Lines & Cumulative Review

Name: _____

Date: _____

FRACTIONS: REVIEW

Number line tip: The space between 0 and 1 is divided into equal parts equal to the denominator. Count "jumps" equal to the numerator.

Part A – Number Line Fractions

Label the marked points on each number line.

1. Each section = $\frac{1}{6}$. Label A, B, C.



A = _____ B = _____ C = _____

2. Plot these fractions on the number line: $\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{2}$



Part B – Mixed Review Sprint

3. $\frac{3}{8} + \frac{1}{4} =$

4. $\frac{5}{6} - \frac{1}{3} =$

5. $\frac{3}{5} \times \frac{2}{3} =$

6. $3\frac{1}{2} + 2\frac{3}{4} =$

7. $5\frac{1}{3} - 1\frac{2}{3} =$

8. $\frac{2}{3}$ of 27 =

Part C – Cumulative: Number Sense + Fractions

9. A bucket holds $3\frac{3}{4}$ gallons. You fill it $\frac{2}{3}$ full. How many gallons are in the bucket?

Answer:

10. A school raised \$48,360. They spent $\frac{3}{8}$ on supplies. How much was spent on supplies? (Hint: find

Week 2 Quiz – Fractions

W2
QUIZ

Topics: Equivalence · Comparing · Add/Subtract · Mixed Numbers · Multiply

Name: _____

Date: _____

Score: _____ / 20

Section 1 – Equivalence & Comparing (5 pts)

1. Fill in the blank: $\frac{3}{4} = \frac{\boxed{}}{28}$

2. Simplify: $\frac{20}{28} = \frac{\boxed{}}{\boxed{}}$

3. Compare: $\frac{5}{8} \boxed{} \frac{3}{5}$

4. Order least to greatest: $\frac{2}{3}, \frac{3}{4}, \frac{5}{8}$

5. **(1 pt)** Convert $6\frac{3}{5}$ to an improper fraction.

Section 2 – Add & Subtract (7 pts)

6. $\frac{1}{3} + \frac{1}{6} =$

7. $\frac{5}{6} - \frac{3}{8} =$

8. $4\frac{3}{4} + 2\frac{2}{3} =$

9. $7\frac{1}{4} - 3\frac{3}{4} =$

10. **(3 pts)** Lena has $5\frac{1}{2}$ feet of ribbon. She uses $2\frac{3}{8}$ feet for one gift and $1\frac{3}{4}$ feet for another. How much ribbon remains?

Answer:

Section 3 – Multiply (5 pts + bonus)

Summer Math Booklet · Grade 4→5 · Week 2 Quiz

11. $\frac{3}{5} \times \frac{5}{9} =$

12. $\frac{3}{4} \times 36 =$

Fractions